

Navigators Technical Documentation

Navigators — Production & Local Deployment Guide

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1. Prerequisites & Environment Setup

1.1 System Requirements

- **Operating System:** macOS (Darwin 24+), Linux (Ubuntu 22.04+ LTS), or Windows with WSL2.
- **Python:** Version 3.11 or higher (developed and verified on Python 3.14).
- **Node.js:** Version 18.0+ LTS (for running client test suites).
- **Git & Git LFS:** Required for repository management and large model weight handling.

1.2 Python Virtual Environment Setup

```
# Clone the repository
git clone https://github.com/Navigators/Navigators.git
cd Navigators

# Create and activate virtual environment
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt
```

2. Database Initialization

The relational SQLite database initializes automatically on first server startup, or can be seeded explicitly:

```
# Initialize schema and default roles/permissions
python3 -c "from src.db.database import init_db; init_db('data/navigators.db')"
```

3. Launching the Backend Server

3.1 Standard Development Server

```
# Run FastAPI with live reloading
uvicorn src.api.server:app --host 127.0.0.1 --port 8000 --reload
```

3.2 Secure Mobile Development Server (HTTPS for Phone Testing)

Mobile browsers require HTTPS to grant access to accelerometer and gyroscope APIs:

```
# Replace with your workstation's local Wi-Fi IPv4 address
python scripts/serve_phone.py --ip 192.168.1.50
```

This command: 1. Generates local development Root CA certificates in `.phone-dev/`. 2. Starts an HTTP bootstrap server on port `8001` to serve the public CA certificate to the phone. 3. Launches the FastAPI backend over TLS on port `8443` (`https://192.168.1.50:8443`).

4. Client PWA Offline Deployment

The client navigation interface is fully static and requires no dynamic server-side rendering: - Serve the `simulator/` directory using any standard static file server (Nginx, Caddy, Cloudflare Pages, or FastAPI static mounts). - Ensure the following MIME types are correctly configured: - `.wasm` → `application/wasm` - `.json` → `application/json` - `.onnx` → `application/octet-stream`

5. Model Rollback & Troubleshooting

Emergency Model Rollback

If a newly promoted model displays abnormal trajectory drift in field evaluation:

```
# Authenticated rollback request
curl -X POST https://127.0.0.1:8000/api/v1/models/rollback \
-H "Authorization: Bearer <team_admin_token>"
```

This immediately reactivates the previous production checkpoint in the database.

Common Troubleshooting Scenarios

1. **Sensors Not Streaming on Android:** Ensure the phone is connected via HTTPS (port 8443), the custom CA certificate is installed, and motion permissions are enabled in Chrome site settings.
2. **WebAssembly SIMD Error:** Verify that the browser version supports WebAssembly SIMD (Chrome 91+, Safari 16.4+, Firefox 89+).

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